

Object oriented Programming language

What is OOP?

OOP (Object-oriented Programming) is a programming language paradigm based on the concept of objects and classes.

---> object

which contain :-

data (attributes or properties)

operations (methods or functions)

+ In oop everything revolves around creating and using classes and objects to organize and manage software.

Why is oop Important in Python?

Python supports both oop and procedural programming but oop offers many advantages especially for larger applications.

Benefit Explanation

code Reusability

classes and inheritance allow reuse of existing code

Modularity

code is divided into logical units making it easier to maintain

Real world modeling - object map closely to real world
maintainability easier to debug update and manage code

Objects

An object is an entity that has attribute and behaviour

For example a "Parrot" is an object It has

Attributes :- name, age, color, etc - -

Behaviours :- dancing singing etc

using variables and methods we can develop the objects

What is class in Python?

A class in Python is a blueprint or template used to create objects

→ class is a keyword using this class keyword we create userdefined classes

→ every userdefined class may contains some attributes & methods

Syntax:-

class userdefined class - Name (predefined - class - Name) :

<attributes>

<methods>

class userdefined - class - Name (object):

<attributes>

<methods>

class userdefined - class - Name:

<attribute>

<methods>

def display():

===

display()

class userdefined class - Name (object):

attributes

def display(self):

===

obj = userdefined - class - Name()

obj.display()

-- It defines the attributes (variables) and methods (functions) that the objects created from the class will have

-- Think of a class as a blueprint for a house

-- you can use the same blueprint to build many houses (objects) each with its own color, size,

→ one class can contain N number of objects

→ when ever creating object for our class then python is instantiated memory for our class and its members

* class by using class keyword we can create
Syntax:-

```
class className :  
    ---
```

```
object = className()
```

What is an object in oop?

An object is an instance of a class in oop

-- It represents a real-world entity with
state (data attribute) and behaviour (methods / functions)

```
class Employee :
```

```
    attributes >
```

```
    /method>
```

```
emp = Employee()
```

```
emp.member
```

```
object.methodname()
```

What is method in python

- A method is a function that belongs to a class
- It defines the behaviour or actions that object created from that class can perform

```
class Employee:
```

```
    salary = 10000
```

```
    city = 'Pune'
```

```
    def display_(self):
```

```
        pass
```

```
    def display2(self, name):
```

```
        pass
```

```
emp = Employee()
```

```
emp.display_()
```

```
emp.display2("ISR")
```

Constructor

```
def __init__(self):
```

```
    ==
```

A constructor is a special method in python we can represent by using `__init__()`

Whenever creating object for our class then this constructor method automatically executing

We have two types of constructor

① Non-Parameterized Constructor

```
def __init__(self):
```

```
    self.a = 10
```

```
    self.b = 20
```

```
obj1 = className()
```

② Parameterized Constructor

```
def __init__(self, a, b):
```

```
    self.a = a
```

```
    self.b = b
```

```
obj1 = className(10, 20)
```

```
obj2 = className(100, 200)
```

Can we create a more than one constructor in a single class?

→ yes, we can create

```
class className:
```

```
    def __init__(self):
```

```
        print('first constructor')
```

```
def __init__(self):
```

```
    print('second constructor')
```

```
obj = class Name()
```